CHAPTER 2:

COMPONENTS OF A BUSINESS INTELLIGENCE ARCHITECTURE

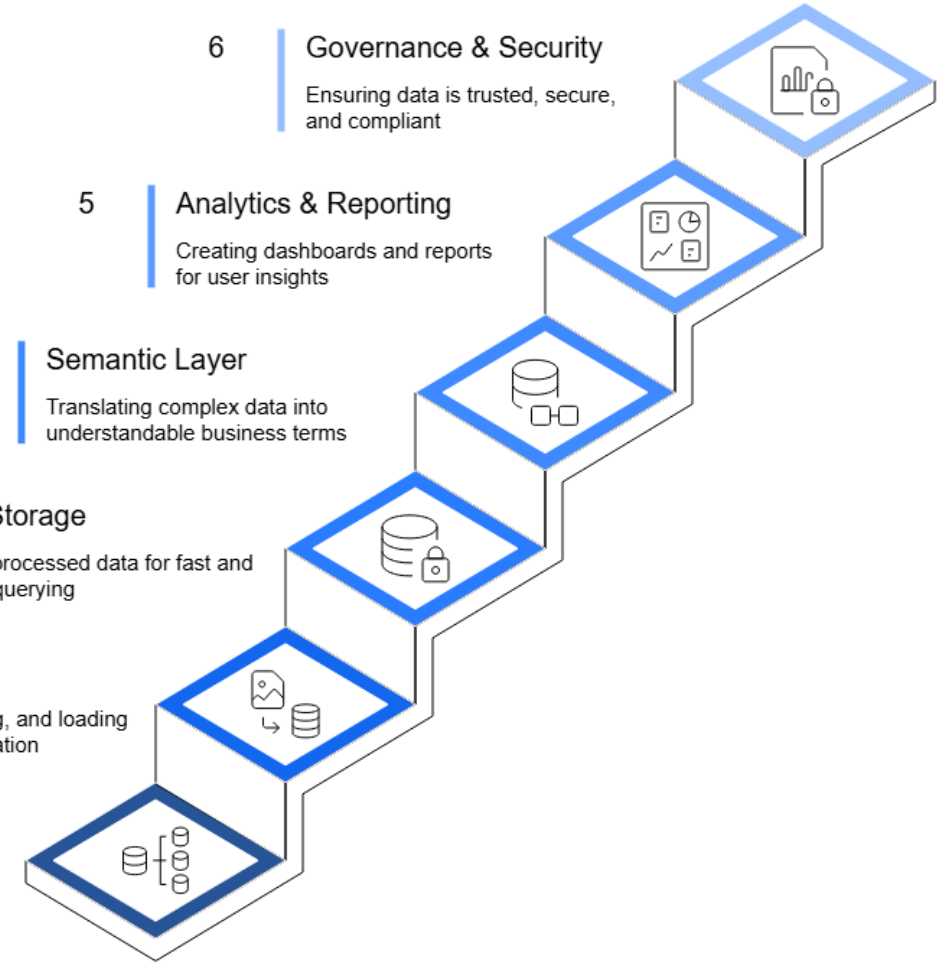
What is Business Intelligence Architecture ?

Business intelligence (BI) architecture is **the framework of technology, processes, and policies that collect, store, and analyze data to support strategic decision-making**. It acts as a **blueprint for transforming raw data into actionable insights,** typically involving data sources, ETL (extraction, transformation, loading), data warehouses, and visualization tools (Examples: Tableau, Power BI).

Components of current BI Architecture

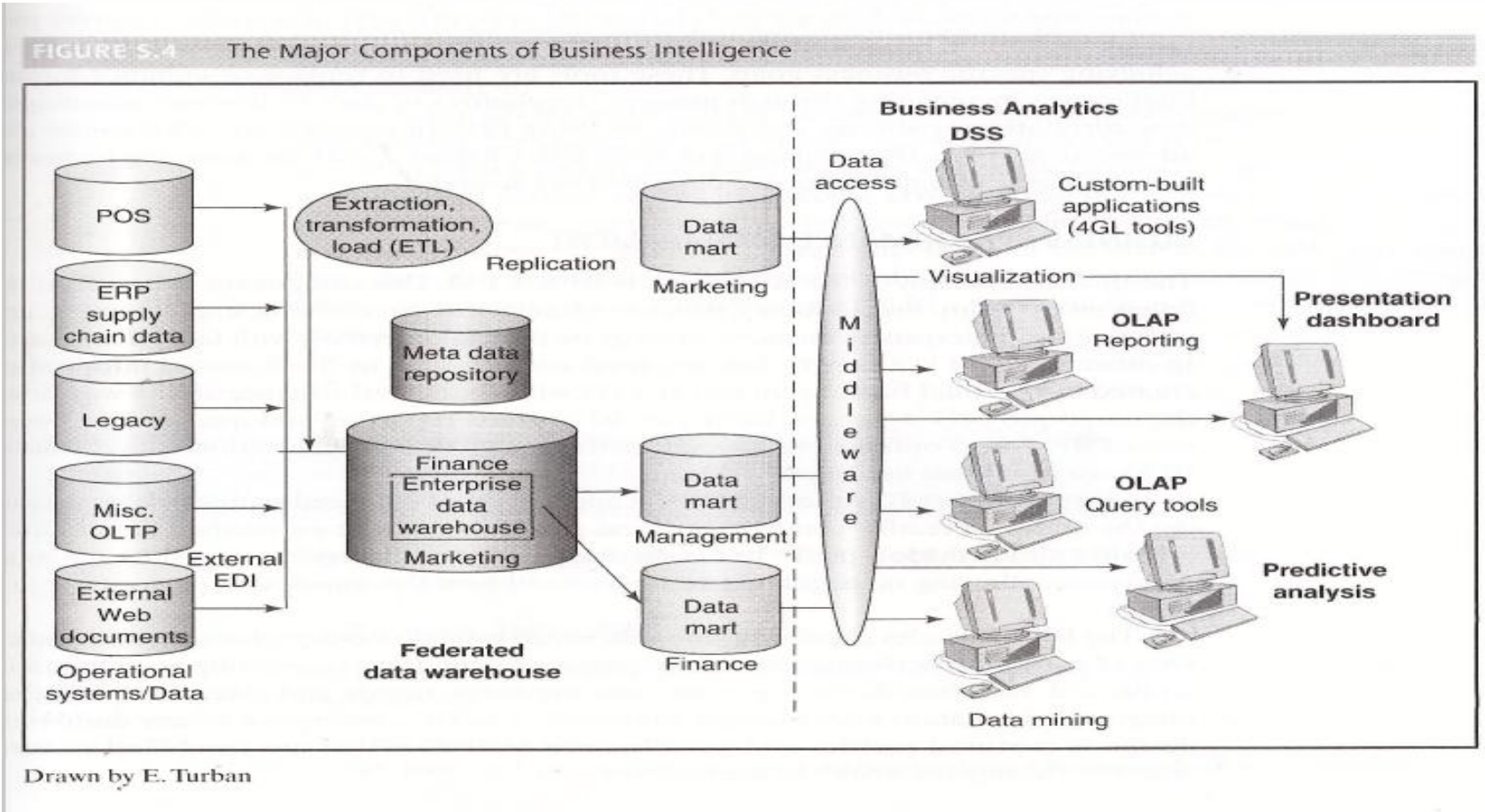
- **Data Sources:** Operational systems, CRM, ERP, and third-party APIs.
- **Data Integration (ETL/ELT):** Tools that move, clean, and organize data.
- **Data Storage:** Data warehouses (example: Snowflakes, Amazon Redshift, Google BigQuery) or data lakes for storing processed data.
- **Semantic Layer:** Maps complex data into business-friendly terms.
- **Analytics and Reporting Layer:** Tools for dashboards, data visualization to analyze data to discover insights.
- **Governance and Security:** Ensures data is trusted, secure, and meets regulatory standards across the BI system

Business Intelligence Architecture



Source: kanerika.com

Major Components in the Business Intelligence Life Cycle



Data Sources

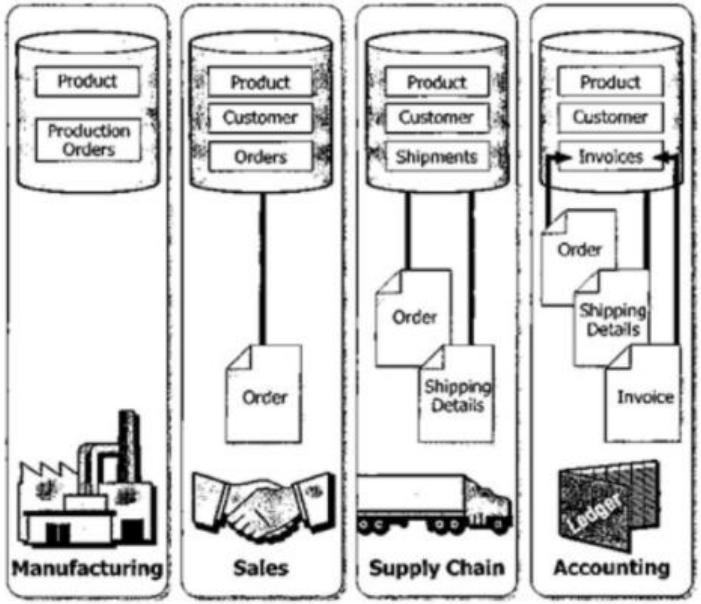
Operational and Source Systems

- Operational systems are the starting points for most quantitative data in a company.

- Operational systems may also be referred to a:
  - ✓ transaction processing systems,
  - ✓ source systems, and
  - ✓ enterprise resource planning (ERP) system.

Operational and Source Systems

- Operational systems record data from operational tasks.



Manufacturing system:

- When a product is produced, the production order is entered in the manufacturing system.
- The quantity of raw material used and the finished product produced are recorded.

Sales system:

- When a customer places an order, the order details are entered in an order entry system.

Supply Chain system:

- When the product is available, the product is shipped and order fulfilment details are entered.

Accounting system:

- Accounting then invoices the customer and collects payments. The invoices and payments may be recorded in an operational system that is different from order entry system.

Operational and Source Systems

- Operational systems record data from operational tasks – cont'd.
  - In each step in this process, users are creating data that can eventually be used for business intelligence.
  - In addition, to complete a task, operational users may need business intelligence.
  - Perhaps in order to accept an order, the product must be available in inventory.
  - As is the case with many online retailers, customers cannot place an order for a product combination (colour, size) that is not available; a report immediately appears with a list of alternative sizes or colours, usually referred to as embedded BI.

Operational and Source Systems

- When BI is integrated with an operational system or supports an operational task, it is referred to as **operational business intelligence**, and the supporting report or nugget of information is called **embedded BI**.
- The operational systems may be custom-developed transaction systems or a purchased package from companies such as:
  - ✓ Oracle (Oracle e-Business Suite, PeopleSoft, J.D. Edwards, Siebel)
  - ✓ SAP
  - ✓ Microsoft (Dynamics GP)
- With custom-developed operational systems or with modules coming from different vendors, data may be manually entered into each system.
- A better approach is to systematically transfer data between the systems or modules.

Operational and Source Systems

- The system should not only can systematically transfer the data, but it must also consistently handle information flows through the process seamlessly.
  - For example – Customer ID that entered in the order system may not be the same as in the accounting system, even though both IDs refer to the same customer systematically transferred but not consistent => data is systematically transferred but the info not consistent.
- ERP systems ensure adherence to standard processes and are broader in scope than custom operational systems of the past.
- From a data perspective, ERPs reduce duplicate data entry and thus improve data quality.
- With an integrated ERP, a common set of reference tables with consistent customer IDs, product code, and chart of accounts are shared across the modules or applications, collectively referred to as **master data**.

Operational and Source Systems

- Within the BI life cycle, the operational systems are the starting point for data you will later want to analyse.
- If you do not capture the data in the operational system, you can't analyse it.
- If the operational system contains errors, those errors will only get compounded when you later aggregate and combine them with other data.

Additional Source Systems

- While much of the data warehouse is populated by internal operational systems, data may also come from additional data sources, such as:
  - Distributors who supply sales and inventory information.
  - Advertisers who supply rates, readership and advertising events.
  - Click-stream data from web logs that show the most frequently viewed products or online shopping cart analysis for partially completed orders.
  - Market prices from external research firms.
  - Social data from Twitter and Facebook where customers may be talking about your products.
  - Machine-generated data, sensors, and radio frequency identification (RFID) that may communicate, for example: Which product is on which rail car and where the rail car is on the track?

Additional Source Systems

- Whether this additional data gets loaded into a central data warehouse will depend on how consistently it can be merged with corporate data, how common is the need for the data, the business value in analysing it, and politics.
- If the data is not physically stored in the data warehouse, it may be integrated with corporate data in a specific data mart or analytics sandbox.
- Disparate data sources may, in some cases, also be accessed or combined within the BI front-end tool, a key feature of many dashboard and visual data discovery products.

Data Integration

Data Transfer: From Operational to Data Warehouse

- BI often involves analysing summary data and combining data from multiple operational systems.
- To facilitate this, data will be extracted from the operational systems and loaded into a data warehouse.
- This process is referred to as **Extract**, **Transform**, and **Load** (**ETL**).
- More recently, some data warehouse teams have changed the order in which they do certain things and call it ELT (Extract, Load, Transform).

Data Transfer: From Operational to Data Warehouse

- The "transform" process of ETL is often the most time-consuming, particularly when multiple disparate systems are involved.
- Inconsistent codes (product ID, customer ID), handling of incomplete data and changing codes to meaningful terms (e.g., 1 = not shipped, 2 = shipped) are all part of the transform process.
- Early data warehouse efforts usually relied on custom-coded ETL, and many still do.

Data Transfer: From Operational to Data Warehouse

- As packaged ETL solutions have come on the market, custom-coded ETL processes have been replaced with purchased ETL solutions.

- Popular solutions for ETL include **Informatica PowerCentre**, **IBM Infosphere DataStage**, **Oracle Data Integrator** and **Microsoft Integration Services** (a component of SQL Server), as well as open-source **Talend**.

Data Transfer: From Operational to Data Warehouse

Why Not Extract Everything?

- In designing a data warehouse, requirements analyst will ask users what they need so that the ETL specialist can configure out what should be extracted from the source systems.

- Because much of BI is unpredictable in nature and users often don't know what they want until they see it, you might ask "why not extract everything?" in the event that you might one day need that data.

Data Transfer: From Operational to Data Warehouse

Why Not Extract Everything?

- There are a number of reasons why all the data may not be extracted:
  - ✓ The time window in which data can be ETL'd (extracted, transformed, and loaded) may be small, especially since many companies and data warehouse serve a global user base.
  - ✓ There can be a negative impact on query performance when too much detailed data is stored in the data warehouse.
  - ✓ Limited time, money and human resources force a prioritization of what data to extract and include in the data warehouse.

Data Transfer: From Operational to Data Warehouse

Enterprise Information Management

- As the data warehouse industry has matured and ETL tools have evolved, this market segment is increasingly referred to as **enterprise information management** (**EIM**).

- EIM includes ETL tools, but also will include:
  - ✓ Data modelling tools,
  - ✓ Data quality,
  - ✓ Data profiling,
  - ✓ Metadata management,
  - ✓ Master data management (MDM)

Data Transfer: From Operational to Data Warehouse

Metadata

- Metadata is data about the data.
- Metadata is similar to an old card file in a library or book details on Amazon.com
- A digital card file in a library (or the book details on Amazon) tell you which category a book belongs to, when it was published, and so on.

Data Transfer: From Operational to Data Warehouse

Metadata (cont'd)

- Metadata may describe such things as:
  - ✓ When the data was extracted from the source system
  - ✓ When the data was loaded into the data warehouse
  - ✓ From which source system an item originated
  - ✓ From which physical table and field in the source system it was extracted
  - ✓ Transformation rules and logic
  - ✓ How something was calculated. For example: revenue = (price x quantity sold) – discounts.
  - ✓ What the item means in a business context (revenue is based on the amount invoiced and does not include returns or bad debts.

Data Transfer: From Operational to Data Warehouse

Metadata (cont'd)

- As you move down the list, the items become much more important to all business users.
- For example: A salesperson may have different definition of revenue than a finance person does.
- As more people use BI, metadata is critical in ensuring a common business terminology and in ensuring users really know what the data means.
- Where the data is extracted from and stored is referred to as **technical metadata**, whereas business definitions and calculations are referred to as **business metadata**.

Data Transfer: From Operational to Data Warehouse

Metadata – Example (technical metadata):

Results - Dynamic Input (19) - Output

19 of 19 Fields | Cell Viewer | Data | Metadata

Record	Name	Type	Size	Source	Description
1	Category	V_String	255	File:C:\Users\User\Desktop\Alteryx_DataPrep Trai...	
2	City	V_String	255	File:C:\Users\User\Desktop\Alteryx_DataPrep Trai...	
3	Country/Region	V_String	255	File:C:\Users\User\Desktop\Alteryx_DataPrep Trai...	
4	Customer Name	V_String	255	File:C:\Users\User\Desktop\Alteryx_DataPrep Trai...	
5	Discount	Double	8	File:C:\Users\User\Desktop\Alteryx_DataPrep Trai...	
6	Order ID	V_String	255	File:C:\Users\User\Desktop\Alteryx_DataPrep Trai...	
7	Postal Code	V_String	255	File:C:\Users\User\Desktop\Alteryx_DataPrep Trai...	
8	Product Name	V_WString	255	File:C:\Users\User\Desktop\Alteryx_DataPrep Trai...	
9	Profit	Double	8	File:C:\Users\User\Desktop\Alteryx_DataPrep Trai...	
10	Quantity	Double	8	File:C:\Users\User\Desktop\Alteryx_DataPrep Trai...	
11	Region	V_String	255	File:C:\Users\User\Desktop\Alteryx_DataPrep Trai...	
12	Sales	Double	8	File:C:\Users\User\Desktop\Alteryx_DataPrep Trai...	
13	Segment	V_String	255	File:C:\Users\User\Desktop\Alteryx_DataPrep Trai...	
14	Ship Mode	V_String	255	File:C:\Users\User\Desktop\Alteryx_DataPrep Trai...	
15	State	V_String	255	File:C:\Users\User\Desktop\Alteryx_DataPrep Trai...	
16	Sub-Category	V_String	255	File:C:\Users\User\Desktop\Alteryx_DataPrep Trai...	
17	OrderDate	Date	10	File:C:\Users\User\Desktop\Alteryx_DataPrep Trai...	
18	ShipDate	Date	10	File:C:\Users\User\Desktop\Alteryx_DataPrep Trai...	
19	Days to Ship Scheduled	Double	8	File:C:\Users\User\Desktop\Alteryx_DataPrep Trai...	

Data Transfer: From Operational to Data Warehouse

Master Data Management

- Master Data Management is the practice of defining and maintaining consistent definitions of business entities (e.g., customer or product) and data about them across multiple IT systems and possibly beyond the enterprise to partnering businesses.

- Master data includes the code and descriptions for: Customer, patient, student, product, charts of accounts, regions, etc.

Data Transfer: From Operational to Data Warehouse

Master Data Management

- Master Data Management is what ensures that the product ID from the product table is ideally the same ID across all the applications.

- This product ID is stored and maintained in one common place so that the relevant operational and business intelligence systems can access and share it.

Data Transfer: From Operational to Data Warehouse

Master Data Management

- In practice, rarely is there a single product ID, for a variety of technical and organizational reasons.

- In this case, master data will include:
  - ✓ the mapping of the different product IDs that really are the same product represented in different systems
  - ✓ hierarchies of how individual products, customers and accounts aggregate and form the dimensions by which you analyse various facts.

Data Transfer: From Operational to Data Warehouse

Do I need a Data Warehouse?

- Many ERP implementations were sold on the promise of delivering business insight, but this is not their main purpose, and ERP alone does not deliver business insight.

- Having a single operational system that ensures consistent business processes and that uses consistent reference data (customer, product code) will make business analysis significantly easier.

Data Transfer: From Operational to Data Warehouse

Do I need a Data Warehouse? (cont'd)

- But, there are a number of fundamental differences between operational systems and an analytics environment.
- The analytics environment may be based on an in-memory engine, data warehouse or data marts.

Data Transfer: From Operational to Data Warehouse

Do I need a Data Warehouse? (cont'd)

- For most companies, you will want a data warehouse separate from the transaction system when:
  - ✓ You need to perform cross-subject or cross-functional analysis, such as products ordered versus inventory on hand.
  - ✓ Such information may exist in two different systems or different modules within an ERP system and are thus combined into the data warehouse.
  - ✓ You want to perform analysis on summary information, aggregated by time (month, quarter) or by some other hierarchy (product groupings). These hierarchies often don't exist in transaction systems and even when they do, running such voluminous queries within a transaction system can slow it to the point of interfering with data entry.
  - ✓ You need consistently fast reporting and analysis times.
  - ✓ Because of their different purpose and design, data warehouse allow for faster queries than operational systems.

Data Storage

Datawarehouse Architecture Approaches

Datawarehouse Architecture Approaches

- **Bill Inmon's paradigm**: Data warehouse is one part of the overall business intelligence system. An enterprise has one data warehouse, and data marts source their information from the data warehouse. In the data warehouse, information is stored in 3rd normal form.

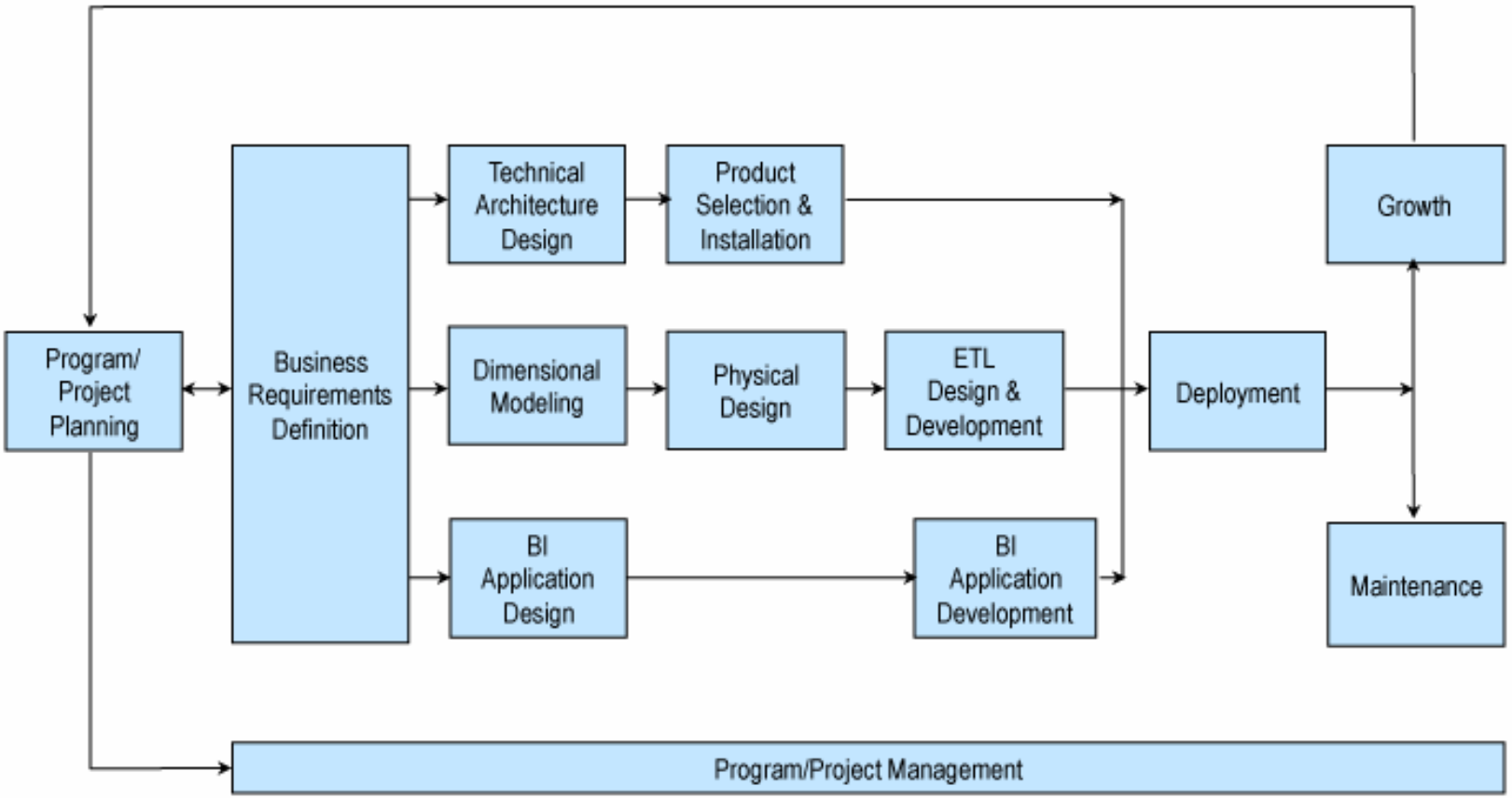
- **Ralph Kimball's paradigm**: Data warehouse is the conglomerate of all data marts within the enterprise. Information is always stored in the dimensional model.

Datawarehouse Architecture Approaches

Feature	Inmon Architecture	Kimball Architecture
Approach	Top-Down	Bottom-Up
Data Model	Normalized (3NF)	Denormalized (Star Schema)
Implementation Time	Longer	Faster
Query Performance	Slower due to normalization	Faster due to denormalization
Scalability	High	Moderate
Flexibility	Less flexible due to centralized design	More flexible with incremental data marts
User-Friendly	Less intuitive for end-users	More intuitive for end-users
Data Redundancy	Minimal	Higher
Complexity	More complex to manage	Easier to manage
Use Case	Large enterprises with comprehensive data integration needs	Smaller to mid-sized organizations with specific business needs

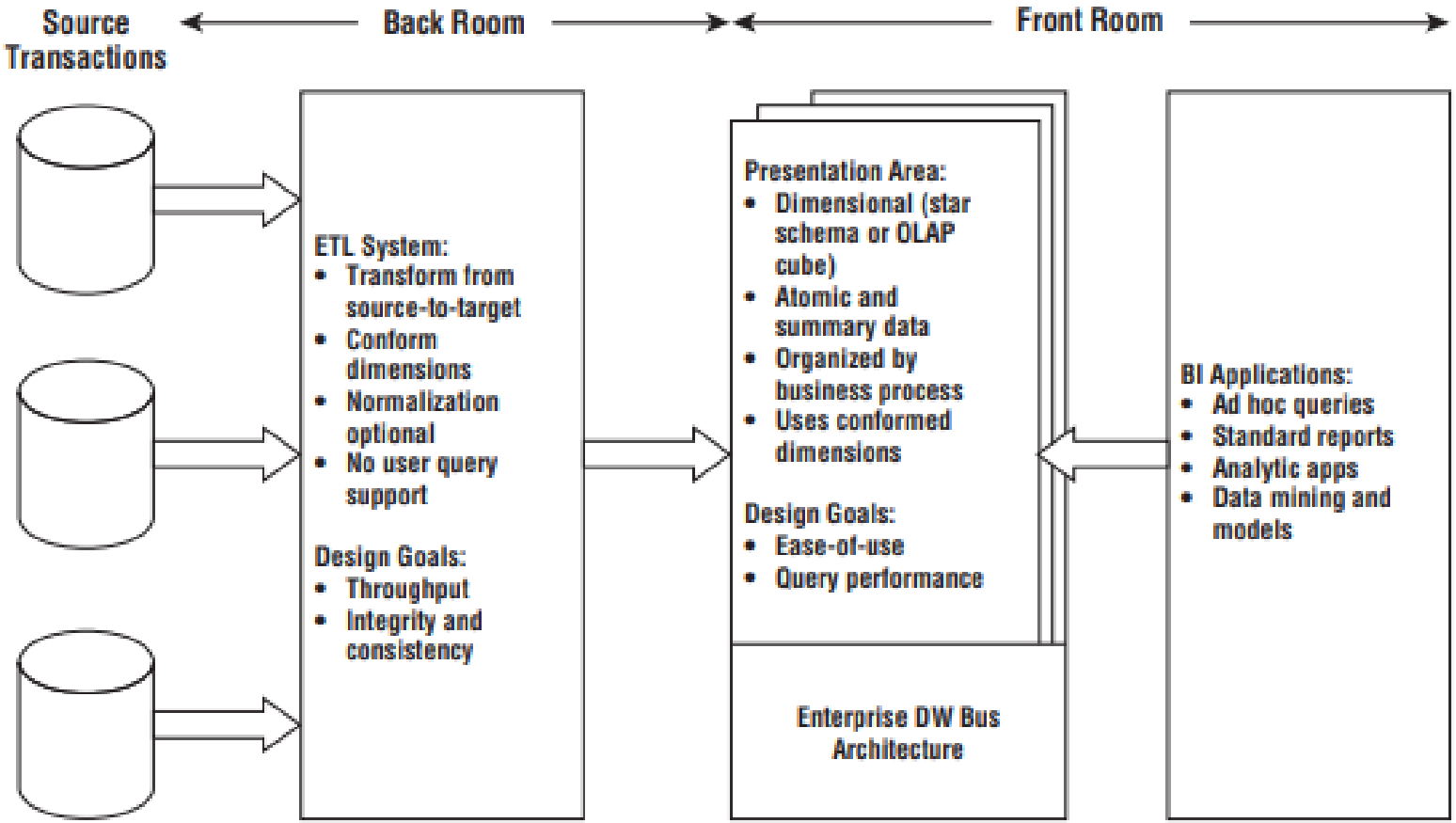
Source: medium.com (Author: Archana Goyal)

Kimball Architecture Approaches



Source: kimballgroup.com; The Data Warehouse Lifecycle Toolkit, Second Edition.

Core Elements of Kimball Architecture



Source: Datawarehouse Toolkit 3rd Edition

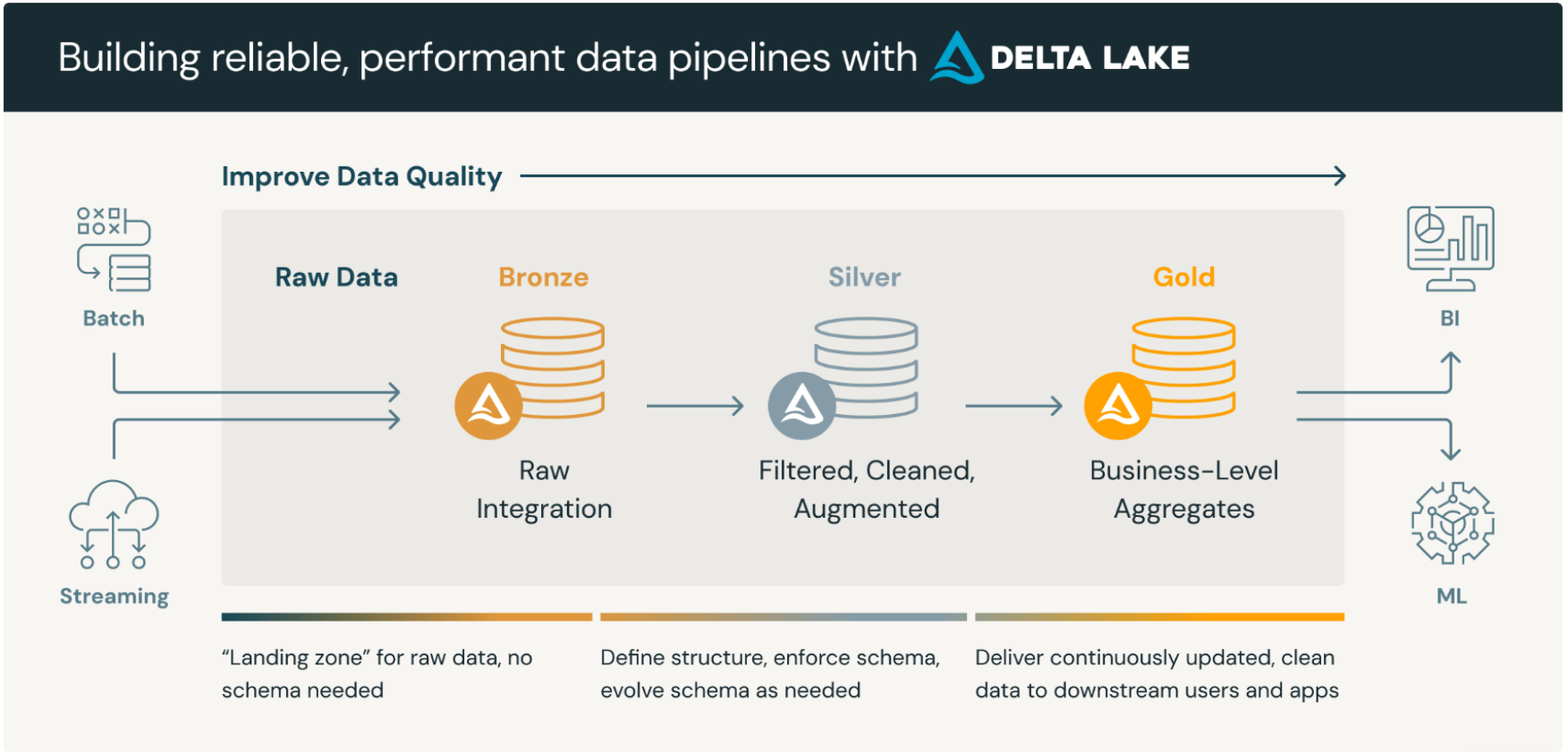
Medallion Architecture (Data Engineering)

- A layered data architecture to improve data quality and usability:
  - **Bronze** Layer: Raw data ingestion from source systems (structured & unstructured).
  - **Silver** Layer: Cleaned and validated data (handling missing values, deduplication).
  - **Gold** Layer: Curated, business-ready data optimized for analytics and reporting.
- Benefits:

a) Improved data quality and governance.

b) Scalable and modular pipeline design.

c) Supports BI, dashboards, and advanced analytics.



Ref: https://www.databricks.com/blog/what-is-medallion-architecture

Data Transfer: From Operational to Data Warehouse

Data Marts

- A data mart:
  - ✓ can be a subset of the data coming from central data warehouse, or
  - ✓ may also be a single subject area populated directly from multiple source systems.
- Whereas a data warehouse is designed to serve the needs of the enterprise, a data mart may serve the needs of a particular business unit, function, process or application.
- Because a data mart is aligned with a particular business requirement, some business may want to skip the data warehouse and build an independent data mart.

Data Transfer: From Operational to Data Warehouse

Data Warehouse Tables

- Within the data warehouse, data is physically stored in individual **tables** within a relational database.

- Your company may use the same relational database software for your ERP system as for your data warehouse (for example: Oracle, Microsoft SQL Server, IBM DB2) or a relational database specifically designed for business intelligence (Teradata, Netezza, Greenplum, Actian ParAccel).

- Experts will deploy a number of different table design approaches to support the diverse business needs, performance requirements and storage constraints.

Data Transfer: From Operational to Data Warehouse

Data Warehouse Tables

- Most data warehouse have two types of tables:
  1) **Fact table** that contains keys into the dimension tables and numeric information to analyse, such as sales, inventory or calls (such facts are often referred to as **measures**).

  2) **Dimension tables** that allow analysis of measures from different perspectives, such as product, time or geography.

Data Transfer: From Operational to Data Warehouse

Data Warehouse Tables: Dimension Tables

- A dimension tables are also referred to as lookup tables or reference tables. It can be broken into more than one table. For example: detailed material IDs may reside in a MATERIAL_ID table.

- The grouping and product hierarchy for the material IDs may reside in a separate table, such as PRODUCT_GROUPING. This type of structure is referred to as a snowflake design and is used in data warehouses that have extremely large dimensions.

- You can think of dimensions as the ways by which you want to analyse facts. For example: ***sales by geography*** or ***sales by product***.

Dimension Tables

- Dimension tables are integral companions to a fact table. The dimension tables contain the textual context associated with a business process measurement event. They describe the "who, what, where, when, how, and why" associated with the event.
- Dimension tables contain descriptive characteristics of business process nouns.

Product Dimension
Product Key (PK)
SKU Number (Natural Key)
Product Description
Brand Name
Category Name
Department Name
Package Type
Package Size
Abrasive Indicator
Weight
Weight Unit of Measure
Storage Type
Shelf Life Type
Shelf Width
Shelf Height
Shelf Depth
...

- Dimension tables often represent hierarchical relationships.. The hierarchical descriptive information is stored redundantly in the spirit of ease of use and query performance. Sample rows from a dimension table with denormalized hierarchies as shown in below Table.

Product Key	Product Description	Brand Name	Category Name
1	PowerAll 20 oz	PowerClean	All Purpose Cleaner
2	PowerAll 32 oz	PowerClean	All Purpose Cleaner
3	PowerAll 48 oz	PowerClean	All Purpose Cleaner
4	PowerAll 64 oz	PowerClean	All Purpose Cleaner
5	ZipAll 20 oz	Zippy	All Purpose Cleaner
6	ZipAll 32 oz	Zippy	All Purpose Cleaner
7	ZipAll 48 oz	Zippy	All Purpose Cleaner
8	Shiny 20 oz	Clean Fast	Glass Cleaner
9	Shiny 32 oz	Clean Fast	Glass Cleaner
10	ZipGlass 20 oz	Zippy	Glass Cleaner
11	ZipGlass 32 oz	Zippy	Glass Cleaner

Source: Datawarehouse Toolkit 3rd Edition

Four-Steps of Dimension Modeling

1. Identify the process being modeled.
2. Determine the grain at which facts will be stored.
3. Choose the dimensions.
4. Identify the numeric measures for the facts.

A multidimensional model consists a fact table and a number of dimension tables

- Fact table - contains measured variables, and identifies them with pointers to dimension tables
- Dimension table - tuples of attributes of dimension

Data Transfer: From Operational to Data Warehouse

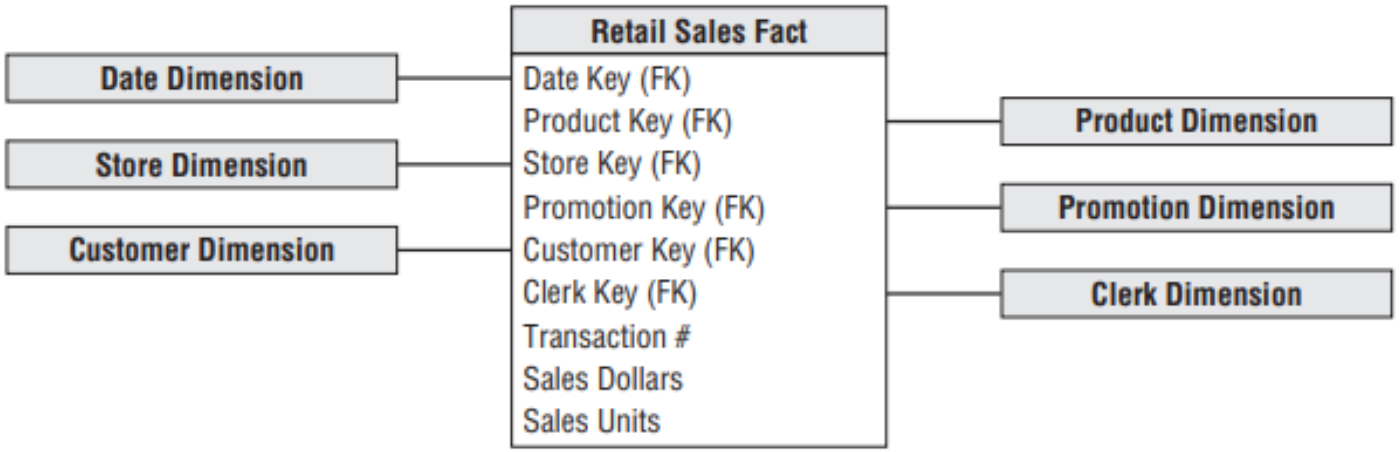
Data Warehouse Tables: Fact/Measures Table

- A fact table can have millions of detailed rows of data commonly referred to as having a “finer granularity”, or can be significantly smaller, containing mainly summary numbers.
- To improve the performance of queries, database designers may choose to create aggregate or summary tables around a fact table such that there may be a DAILY_SALES_FACT table, MONTHLY_SALES_FACT table, and YEARLY_SALES_FACT table.
- One fact table together with its associated dimension tables is referred to as a **star schema**.

Facts & Dimension Tables

- Each business process is represented by a dimensional model that consists of a fact table containing the event's numeric measurements surrounded by dimension tables.

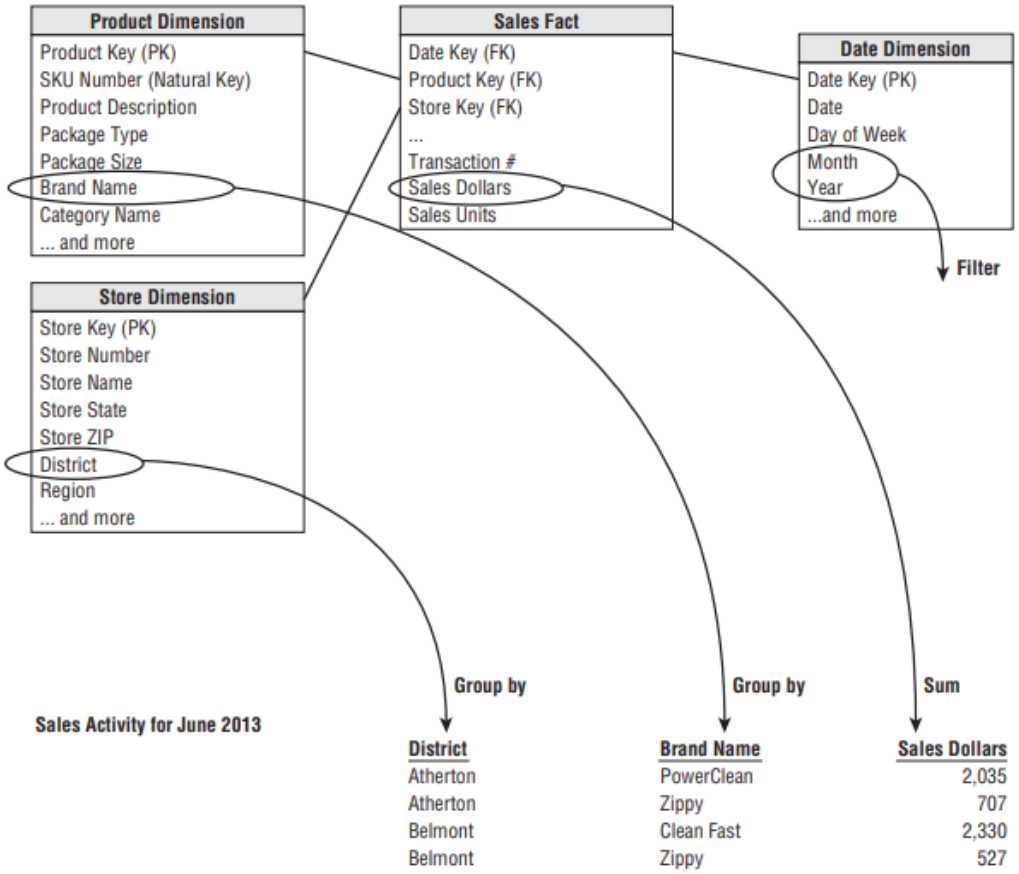
- Every dimension is equivalent; all dimensions are symmetrically equal entry points into the fact table.



Source: Datawarehouse Toolkit 3rd Edition

Facts & Dimension Tables

- Dimensional attributes and facts form a simple report as shown in the example.
- Dimension attributes supply the report filters and labeling, whereas the fact tables supply the report's numeric values.

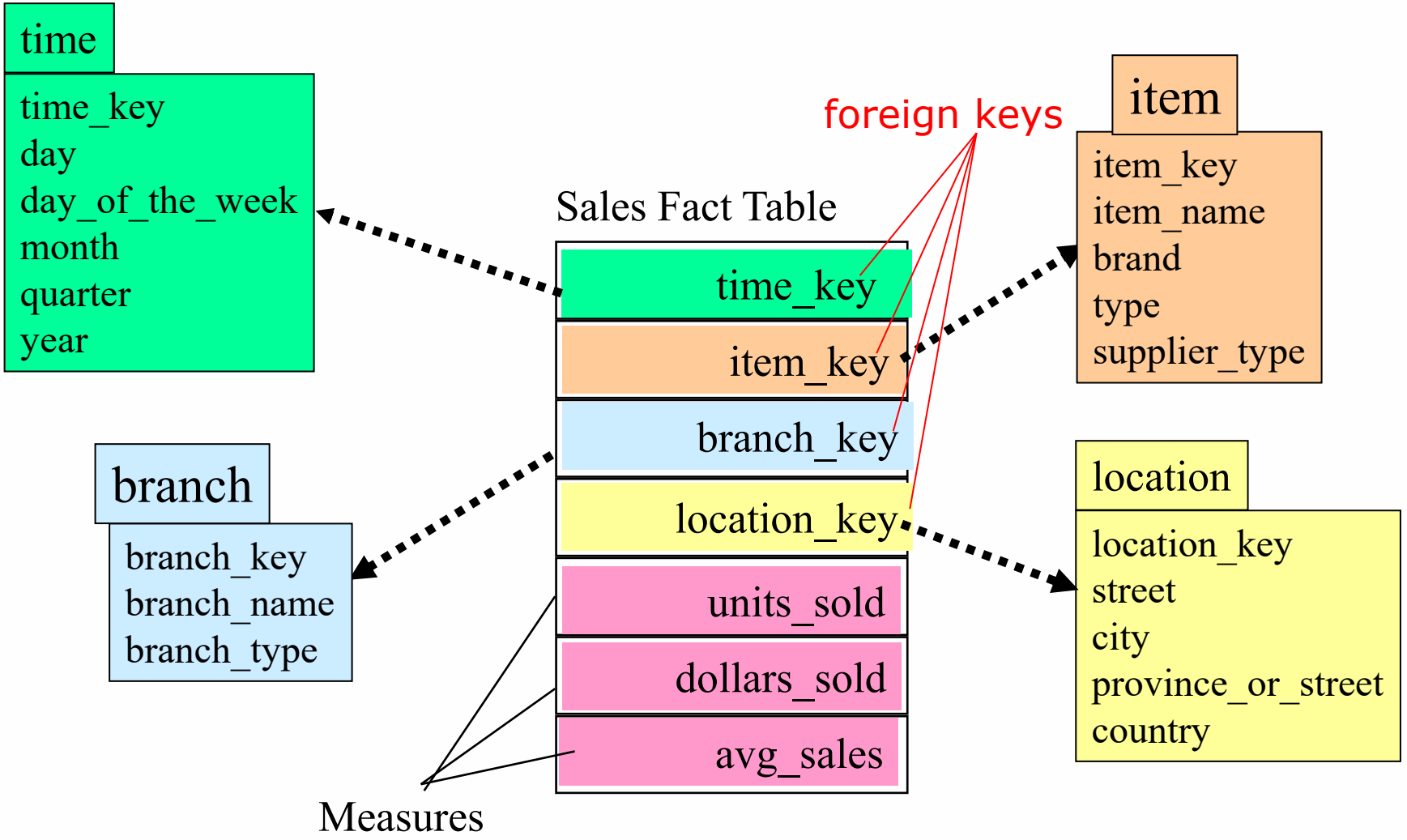


Source: Datawarehouse Toolkit 3rd Edition

Conceptual Modeling of Data Warehouse

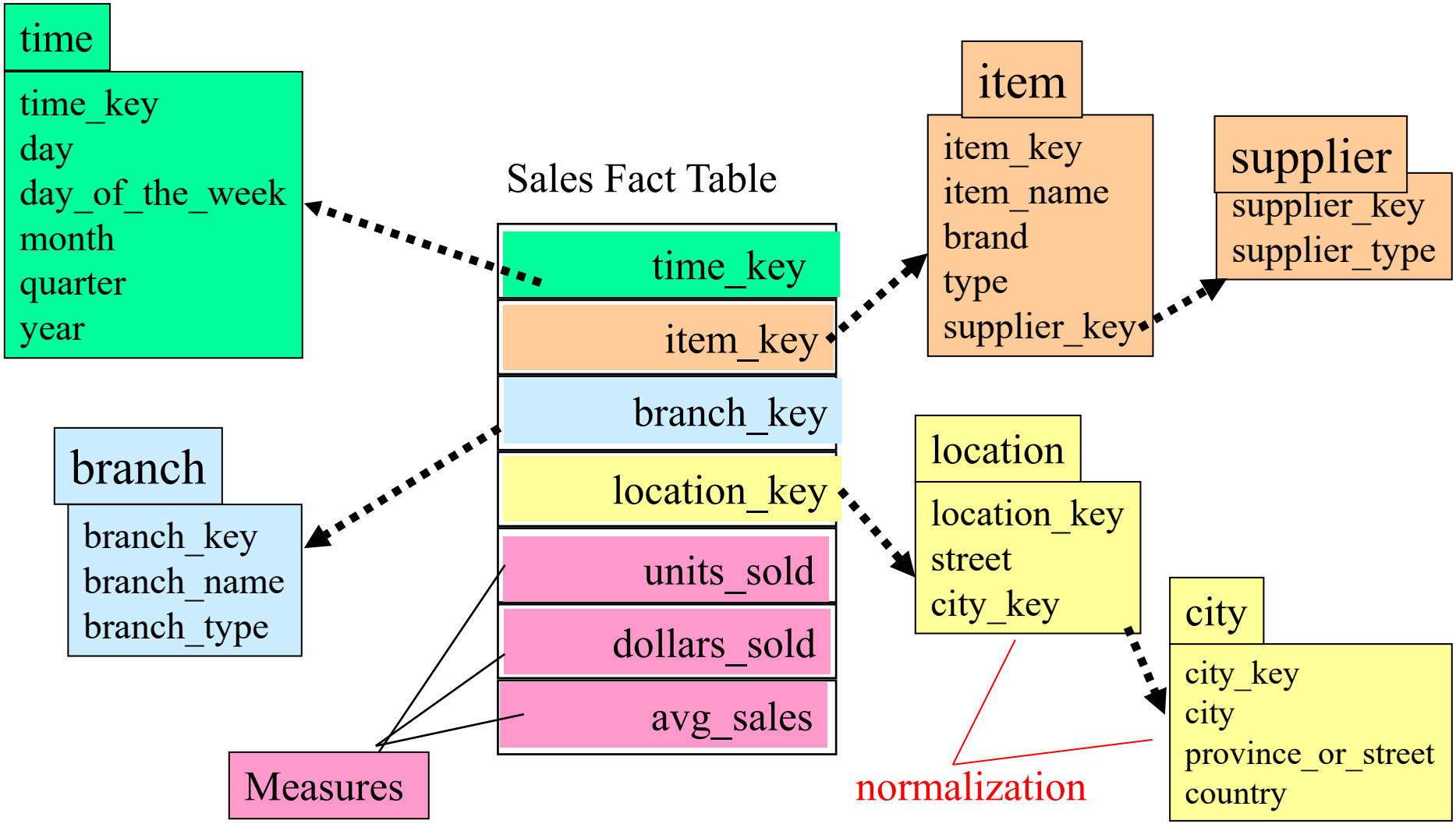
- The ER model is used for relational database design. For data warehouse design we need a concise, subject-oriented schema that facilitates data analysis.
- Modeling data warehouses: dimensions & measures , typical examples:
  - Star schema: A fact table in the middle connected to a set of dimension tables
  - Snowflake schema: A refinement of star schema where some dimensional hierarchy is normalized into a set of smaller dimension tables, forming a shape similar to snowflake
  - Fact constellations: Multiple fact tables share dimension tables, viewed as a collection of stars, therefore called galaxy schema or fact constellation

Star Schema Example



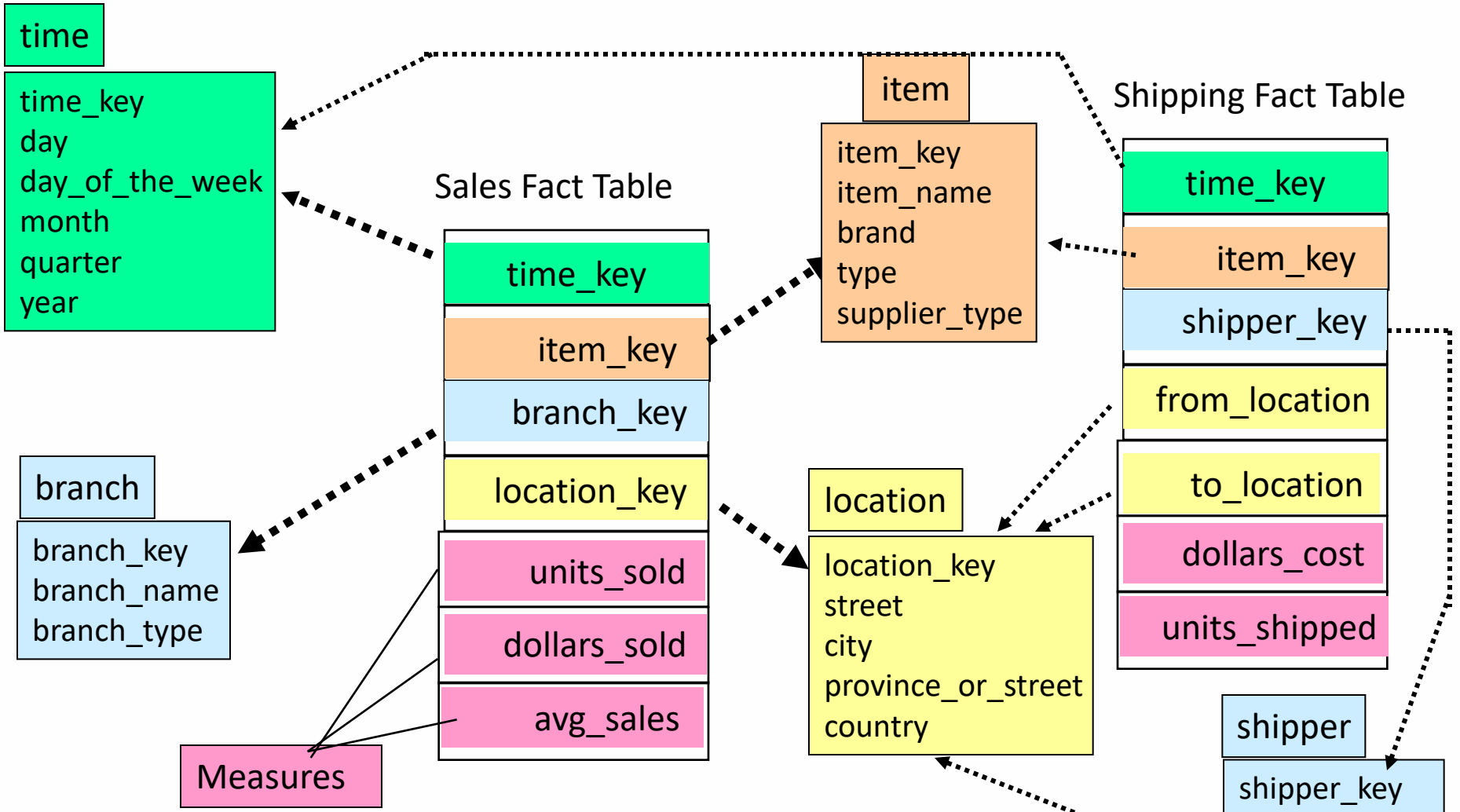
Source: https://www.slideshare.net/slideshow/data-warehouse-21-other-varients-of-star-schema/232750981#3

Snowflake Schema Example



Source: https://www.slideshare.net/slideshow/data-warehouse-21-other-varients-of-star-schema/232750981#3

Fact Constellation Example



Source: https://www.slideshare.net/slideshow/data-warehouse-21-other-varients-of-star-schema/232750981#3

Data Transfer: From Operational to Data Warehouse

Data Storing

- In some respects, business users may not care about how the data is physically stored, whether in the data warehouse or in the transaction system.

- A business view in the BI tool will often hide such technical issues. However, the better that business users can define requirements in advance, the better that data modelers might be able to store data in a way that facilitates the analysis.

Data Transfer: From Operational to Data Warehouse

Data Storing (cont'd)

- For example: If a user wants to analyse something like staffing levels versus sales performance and these two subjects exist in different fact tables and data marts, such analysis can be challenge with certain BI tools.

- If users want to routinely analyse these two different subject areas together, then the data modeller may ultimately decide to store them in one common fact table.

The Data Warehouse & Technology Platform

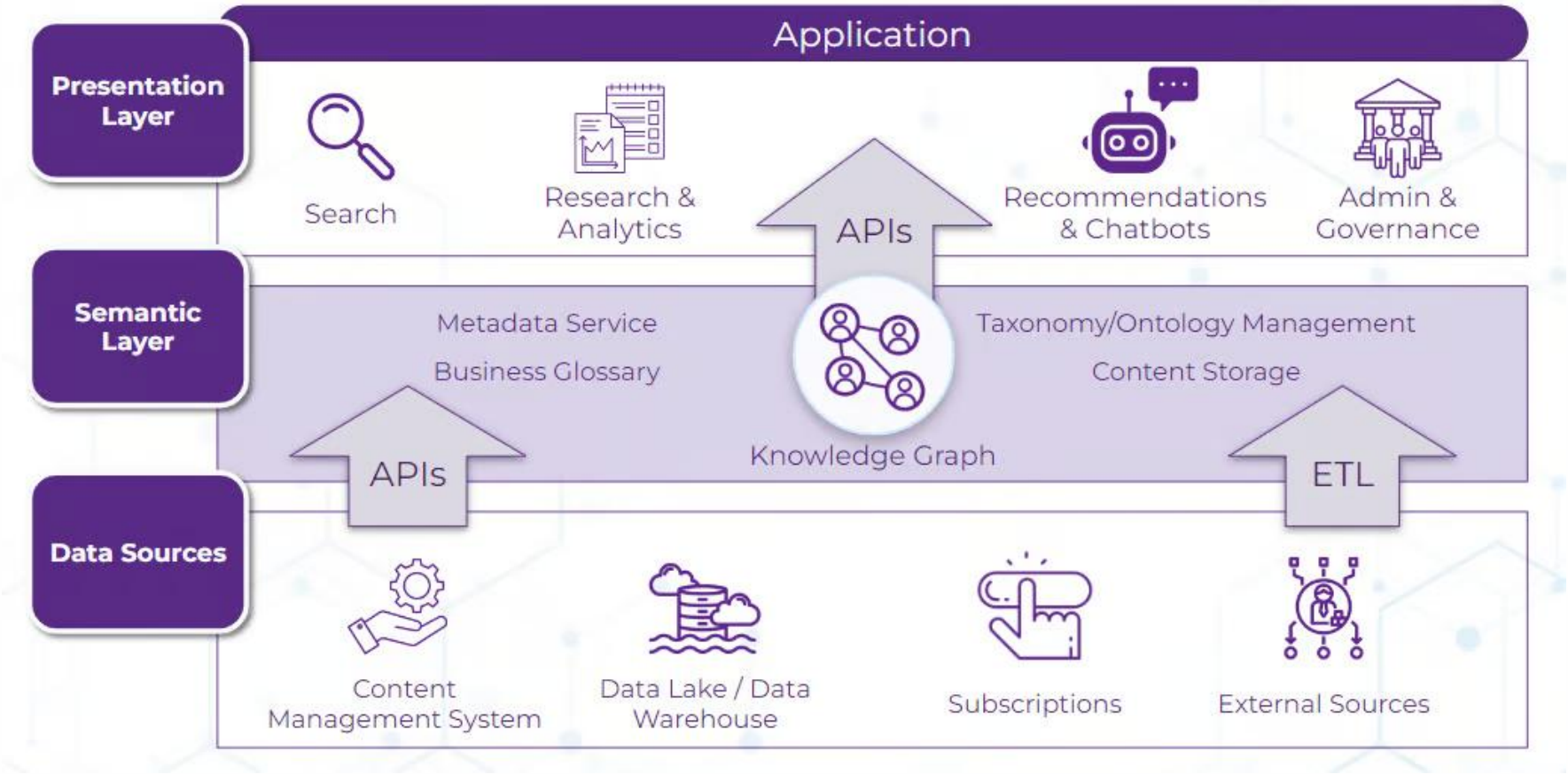
- To drive a car, you need roads, highways and a transportation infrastructure. Similarly, just as in a BI environment, a number of servers and networks may be involved:
  - ✓ The server(s) on which the relational database management system (RDBMS) is running.
  - ✓ The server(s) that run the ETL software and processes.
  - ✓ The web server(s) that provide the entry point into the BI environment.
  - ✓ The BI server(s) that process queries, dashboard and reports.

Semantic Layer

What is Semantic Layer ?

- A semantic layer is a piece of enterprise data architecture designed to simplify interactions between complex data storage systems and business users.
- Highly skilled data engineers understand raw data, but most business users do not have the deep technical expertise needed to extract insights easily from raw data.
- The semantic layer provides a user-friendly interface that converts that data into meaningful business terms. It allows users to focus on analyzing data rather than on the technicalities of data retrieval.

Main Components of Semantic Layer



Source: [Enterprise Knowledge]

Analytics & Reporting

The Data Warehouse & Technology Platform

Analytic Appliances

- Data warehouse appliances combine the **server**, **database** and **data storage** into one system.

- Leading data warehouse appliance vendors include IBM Netezza, EMC Greenplum, Teradata, HP Vertica, Oracle Exadata, SAP Sybase IQ and 1O1Odata. SAP Hana is an in-memory database that runs on hardware manufactured by a number of hardware vendors.

- The promise of an appliance is a complete, optimized solution that delivers better performance at a lower cost of ownership than if a company were to purchase and install these individual components.

The Data Warehouse & Technology Platform

Cloud Business Intelligence

- Initially, cloud BI was restricted to Software as a Service (SaaS) solutions.
- With an SaaS model, customers simply subscribe to a solution and typically upload their data to the cloud.
- A third-party vendor hosts the technical infrastructure that customers then access via the web.

The Data Warehouse & Technology Platform

Cloud Business Intelligence

- As the cloud has matured, there are a number of variations in how it can be used for a BI implementation:

1) Infrastructure as a Service (IaaS) is when a third-party provisions virtualized servers, storage and network resources to run databases and applications.

- ✓ Among the provider of Iaas: GoGrid, Savis, Backspace and Amazon Web Service (AWS)
- ✓ With IaaS, you pay as you go for the hardware, storage or computing power and will then buy and deploy your own software to build a data warehouse and BI solution.

2) Platform as a Service (PaaS) is when a third-party provides optimized software on top of IaaS that can be used to build applications. The preconfigured software may include application servers and database software, as well as the BI software.

3) Software as a Service (SaaS) includes the infrastructure and applications that can be directly consumed by business users.

- ✓ Some SaaS vendors will use IaaS providers.
- ✓ Most often, the data is replicated from an on-premise environment and loaded in the BI SaaS database, but a few allow a reach through to on-premise data.

The Data Warehouse & Technology Platform

Cloud Business Intelligence

- Example:
  - *Netflix* streaming business unit uses AWS for delivering content to viewers, so it makes sense that the data warehouse is also housed in the cloud using Amazon's Storage Center (S3).
  - *FlightStats* has to serve up flight updates to 5 million users a month, so it uses AWS for its infrastructure.
  - *Huntington Bank* chose to MicroStrategy Cloud for a new commercial load dashboard.

The Important of Business Intelligence Tools

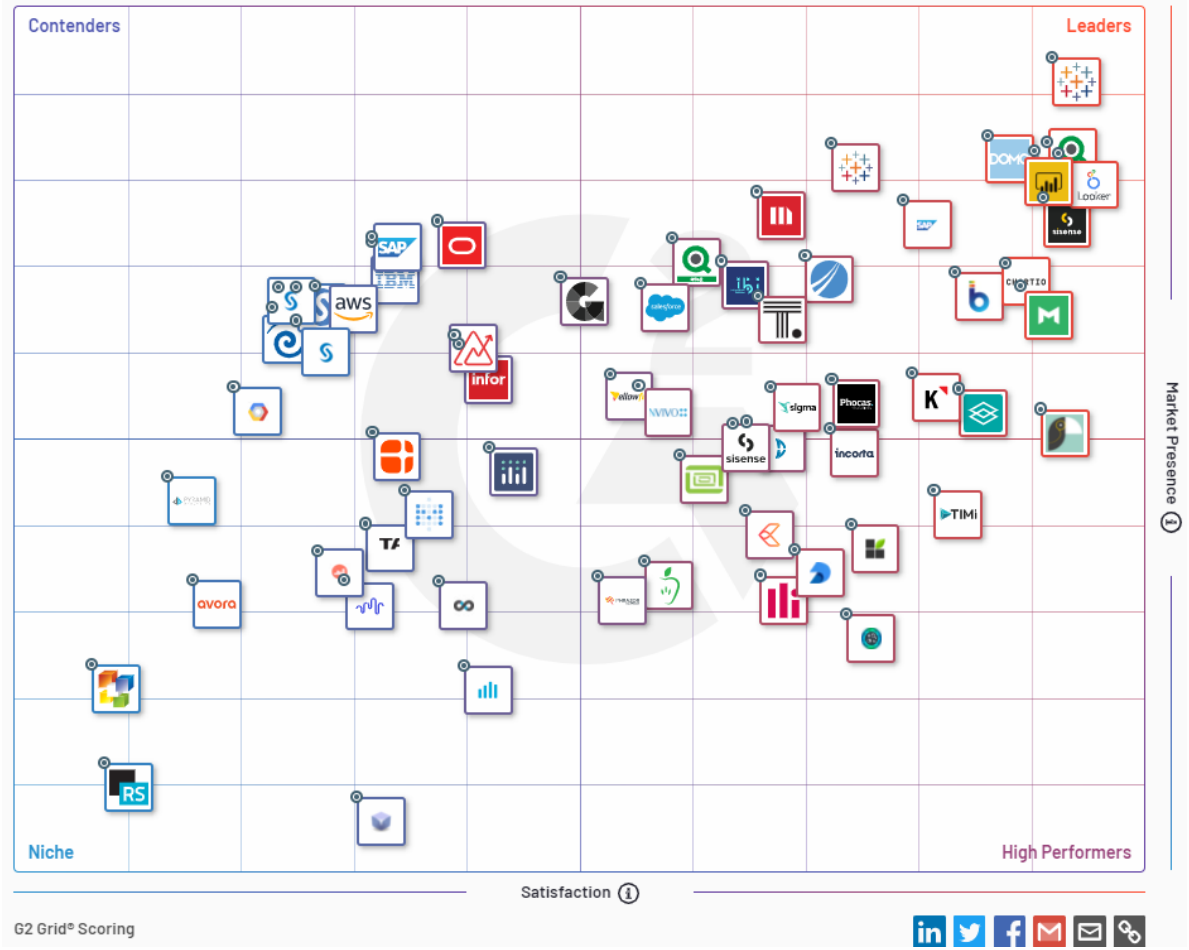
Analytics Platforms

- Analytics platforms, sometimes known as business intelligence (BI) platforms, provide a tool set for businesses to absorb, organize, discover, and analyze data to reveal actionable insights that can help improve decision-making and inform business strategy.
- Enable companies to gain visibility into their data through data integration, cleansing, blending, enrichment, discovery, and more. These tools are robust systems that sometimes require IT and data science skills to access and decipher company data through custom queries.
- Analytics platforms offer a comprehensive look into a company's data by pulling from both structured and unstructured data sources through a series of detailed queries.

(ref: https://www.g2.com/categories/analytics-platforms)

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Analytics Platforms



(ref: https://www.g2.com/categories/analytics-platforms#grid)

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Analytics Platforms

Figure 1: Magic Quadrant for Analytics and Business Intelligence Platforms



Source: Gartner (February 2021)

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Analytics Platforms



Tableau Desktop



Microsoft Power BI Desktop



Qlik Sense

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BI Application Areas:

- BI/Analytics can be applied in all “businesses” (industries, functional areas, or domains) to drive “business” performance.
- Companies (for profit) and financially related :
  - ✓ Retail, manufacture, real-estate, financial, sports, media, advertising, entertainment, healthcare, publication, energy, etc.
- Public (non-profit) :
  - ✓ Organization, institution, association, community, etc.
- Government:
  - ✓ citizen service, city planning, crime, immigration, etc.
- Personal:
  - ✓ personal health, exercise, learning, eating, power consumption, etc.

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BI Application - Example:

- Human Resource/Capital:
  - ✓ HR Analytics
  - ✓ Talent management
- Project and program management
- Power and energy management
- Healthcare management
- IT management:
  - ✓ Web analytics
  - ✓ App analytics
  - ✓ Security management

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BI Application - Example:

- Business management:
  - ✓ Strategic planning
  - ✓ Performance management
  - ✓ Process intelligence
  - ✓ Competitive intelligence
- Marketing and sales:
  - ✓ Customer Relation Management (CRM)
  - ✓ Customer behavior analysis
  - ✓ Targeted marketing and sales strategies
  - ✓ Customer profiling
  - ✓ Campaign management
  - ✓ Inventory management
- Supply-Chain and Logistics:
  - ✓ Supplier and vendor management
  - ✓ Shipping and inventory control

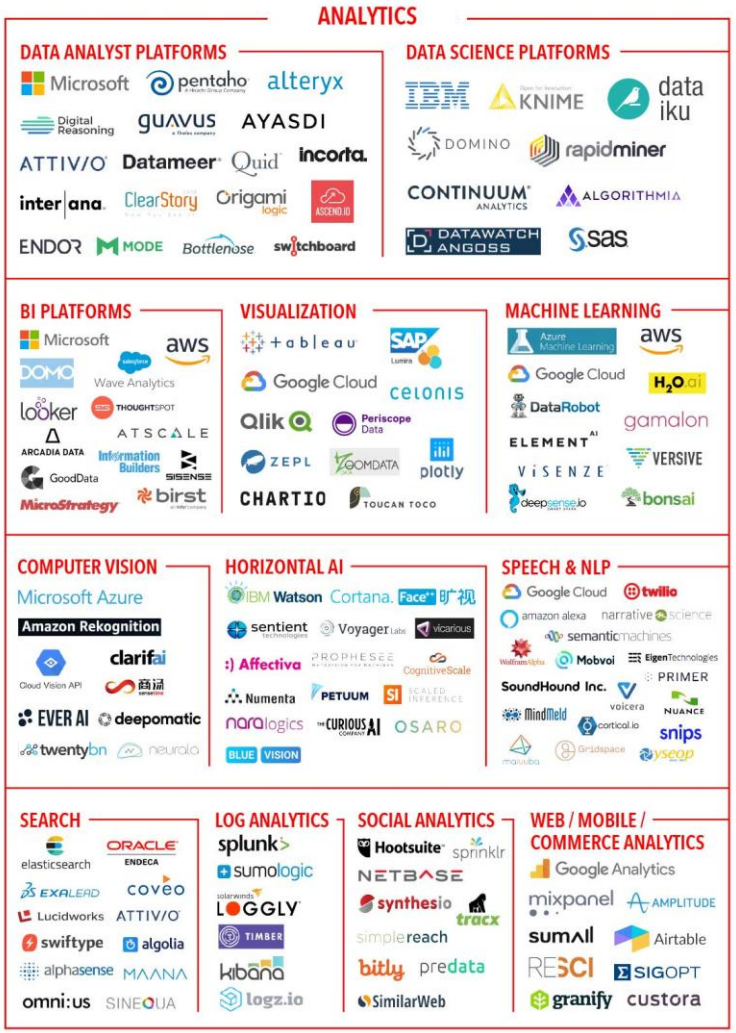
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BI Application - Example:

- Insurance
- Government:
  - ✓ City planning
  - ✓ Traffic management
  - ✓ Urban Analytics
  - ✓ Power usage
- Education
  - ✓ Learning analytics
  - ✓ Student engagement and success
  - ✓ Institutional effectiveness
- Social analytics
- Sports and games analytics

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BI Application - Example:



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Business Analytics:

- There are many software tools for users to create on-demand reports and queries and analyze data. (common name OLAP online analytical processing)
- Users could analyse different dimensions of data and trends. Business users easily identify performance trends by using trend analysis and graphic tools There are three groups categories of tools:

➢ Reporting and queries – we have to do with all types of queries, discovery of information,multidimensional vew,drilldown to details and so on

➢ Advanced Analytics – include many statistical, finacial, mathematical and other models used in analyzing data and information

➢ Data, Text and Web Mining – data mining is a process of searching for unknown or nonobvious relationship or information in large databases using intelligent tools (neural computing or advanced statistical methods) on quantitative data, text, or web data.

Ref: https://slideplayer.com/slide/12222862/72/images/1/Business+Intelligence+Systems.jpg

The Important of Business Intelligence Tools

How BI Provide Value?

- Business Intelligence cuts across all functions and all industries.
- BI touches everyone in a company and beyond to customers, suppliers and with public data to citizens.
- BI can only provide value when it is used effectively by people.
- There is correlation between the effective use of BI and company performance.
- However, simply having better access to data does not improve performance; the difference is in what companies do with the data.

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The Important of Business Intelligence Tools

BI for Management and Control

- In its most basic sense, Business Intelligence provides managers information to know what's going on in the business.
- Without BI, managers may talk about how they are "flying blind" with no insight until quarterly financial numbers are published.
- With BI, information is accessible on a timelier (faster) and more flexible basis to provide a view of:
  - Sales in various regions and by various product lines
  - Expenses compared to budget
  - Warehouse inventory for a particular product or raw materials.

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BI for Management and Control – cont'd

- When any particular metric is not where it should be, BI allows users to explore the underlying details to determine why metrics are off target and to take action to improve the situation.
- In the past, if manager monitored the business via paper-based reports or a fixed screen in a transaction system, they had no flexibility to explore why the business was operating a certain way.
- For example, many companies use BI to monitor expenses to ensure costs do not exceed budgets.
- Rather than waiting until the close of the quarter to discover that excessive expenses have reduced profitability, timely access to expense data allows managers first to identify which business unit is over budget and then to take immediate steps to reduce overtime pay or travel expenses, or to defer purchases, for example.

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The Important of Business Intelligence Tools

BI for Improving Performance

- Use effectively, BI allows organizations to improve performance (efficiency).
- Business performance is measured by a number of financial indicators, such as:
  - ✓ Revenue
  - ✓ Margin
  - ✓ Profitability
  - ✓ Cost to serve
- In marketing, performance gains may be achieved by improving response rates for particular campaigns by identifying characteristics of more responsive customers. Eliminating ineffective campaigns saves companies millions of dollars each year.

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BI for Improving Performance – cont'd

- BI allows companies to boost revenues by cross-selling products to existing customers.
- Accounting personnel may use BI to reduce the aging of accounts receivable by identifying late-paying customers.
- In manufacturing, BI can facilitate a gap analysis to understand why certain plants operate more efficiently than others.

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Operational Business Intelligence

- Operational business intelligence most differs from BI for management and control purposes in both the level of details required and in the timeliness of the data.
- Operational BI may involve accessing a transaction system directly or through a data warehouse that is updated in near real time multiple times throughout the day.
- Business intelligence for management and control purposes may also be in near real time, but can also be based on weekly or monthly data.

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Business Intelligence for Process Improvement

- The operations of a business are made up of dozens of individual process.
- BI may support the decisions individuals make in every step of a process. It also may be used to help streamline a process by:
  - measuring how long sub processes take,
  - identifying areas for improvement.
- For example:
  - manufacturing-to-shipment is one process.
  - In the absence of BI, a company may only realize there is a problem when a customer complains: “My order is late” or “I can get that product faster from your competitor”
  - By analysing the inputs, the time and the outputs for each step of the process, BI can help identify the process bottlenecks.

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Business Intelligence to Improve Customer Service

- The quality of customer service eventually manifests itself in the financial of a company.
- BI can help companies deliver high customer service levels by providing timely order processing, loan approvals, problem handling, and so on.
- For example:
  - Whirlpool uses BI to monitor its warranty program to understand root causes for warranty problems and improve customer satisfaction with its products.
  - United Airlines uses BI to monitor how full business-class cabins are and to ensure its most valued customers receive complementary upgrade when space permits.
  - Netflix tracks how often a customer gets their first-choice DVD.

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Business Intelligence for Discovering New Business Opportunities

- Business intelligence helps business assess and uncover new business opportunities by exploring data and testing theories.

- For example:
  - Companies use data to understand the value proposition of pursuing joint ventures and mergers and acquisitions.
  - A hospitality company uses business intelligence to explore hotel capacity rates as a way of developing the time-share business.

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Technology Changes Enabling Business Intelligence – cont'd

- While data has gotten bigger, ensuring a fast time to insight has gotten harder.
- Researchers have noted that when decision-makers are presented with more data, decision-making is slowed.
- We want to make a perfect decision and to be sure we have assessed every relevant input.
- When BI is deployed effectively, all that data becomes a strategic asset to be exploited.

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At the Speed of Thought

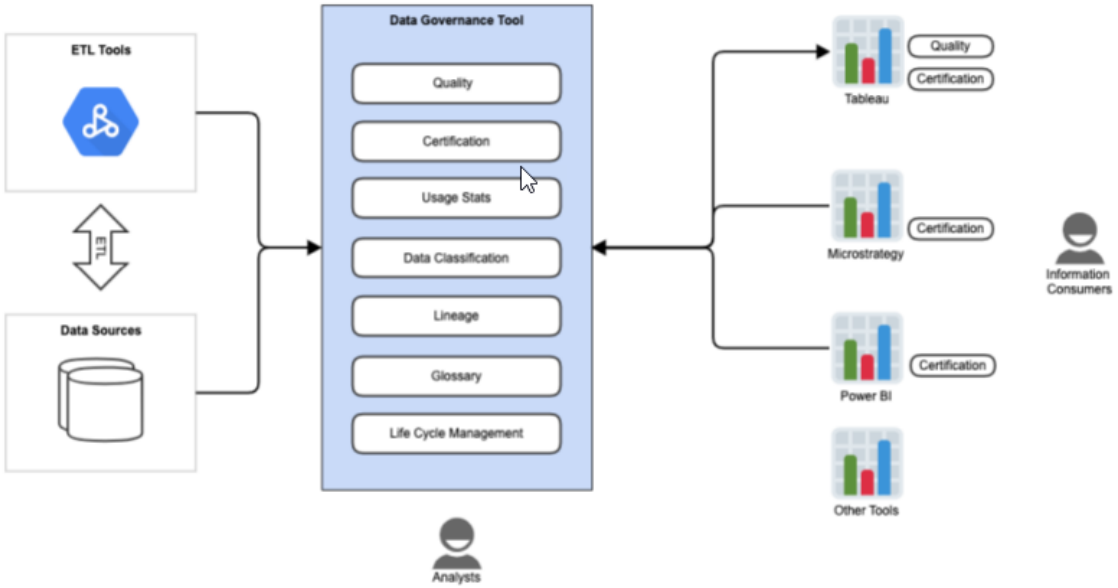
- It might seem that with the explosion of data, accessing more data would get slower.
- Yet computer processing power and addressable memory have increased to the point that accessing large volumes of data can now be done at the speed of thought.
- Twenty years ago, you might have waited a month for a complex, printed report that ran on a mainframe computer for days.
- Ten years ago, the same report might have taken hours, a marginal improvement.
- Today, the same report may run in subseconds on a purpose-built BI appliance and be delivered to a smartphone.

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Governance & Security

Data Governance vs BI Governance

- **Data Governance** is the set of processes and technologies that are used to ensure effective management and utilization of data.
- Analyst and Data Stewards in an organization use Data Governance Tools to simultaneously enforce corporate governance policies and to promote correct usage of data.



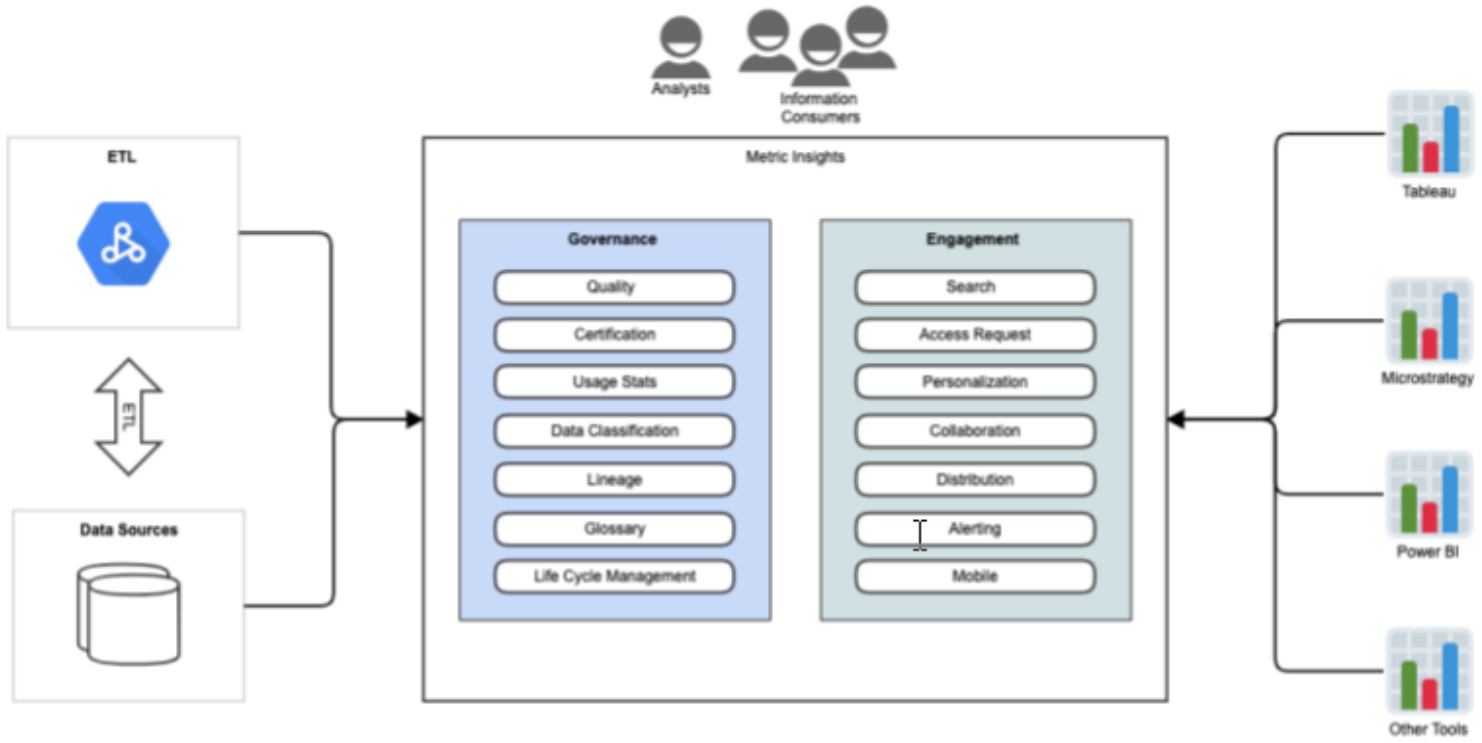
Ref: https://www.metricinsights.com/wp-content/uploads/2021/06/Data-Governance-vs-BI-Governance-Whitepaper.pdf

Data Governance vs BI Governance

- **BI Governance** extends the scope of traditional data governance to reach all information consumers in an organization by leveraging governance metadata to drive user engagement and improve overall data literacy.
- A BI portal:
  - ✓ can be used as the foundation of effective BI Governance as it acts as the single pane of glass through which everyone on the enterprise accesses analytics.
  - ✓ Create a tightly-knit relationship between analytics consumption and governance.
  - ✓ Made the governance metadata broadly available to all users, not limited to a number of analyst only.

Ref: https://www.metricinsights.com/wp-content/uploads/2021/06/Data-Governance-vs-BI-Governance-Whitepaper.pdf

Data Governance vs BI Governance



Ref: https://www.metricinsights.com/wp-content/uploads/2021/06/Data-Governance-vs-BI-Governance-Whitepaper.pdf

Data Governance vs BI Governance

- Traditional data governance focuses only on the data and stops at the shore of visualization. This governance approach is inadequate because most information consumed in an organization is through BI and reporting tools.

- Effective BI governance that encompasses both data and visualization can be achieved by leveraging a BI Portal.

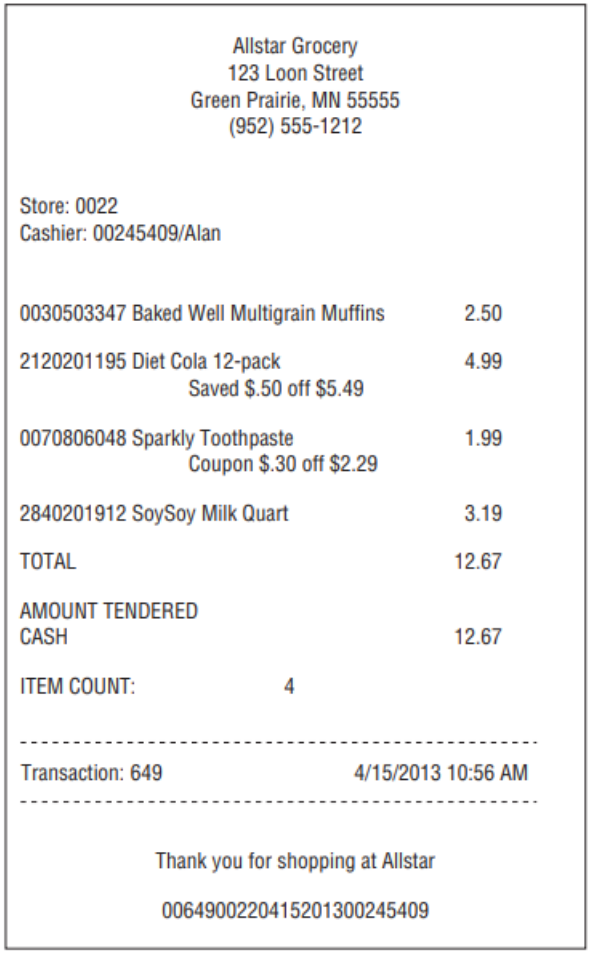
- A BI portal can be used as a stand-alone governance solution or it can be deployed alongside Data Catalog tools.

Ref: https://www.metricinsights.com/wp-content/uploads/2021/06/Data-Governance-vs-BI-Governance-Whitepaper.pdf

Case Study Exercise

Retail Case Study

Imagine you work in the headquarters of a large grocery chain. The business has 100 grocery stores spread across five states. Each store has a full complement of departments, including grocery, frozen foods, dairy, meat, produce, bakery, floral, and health/beauty aids. Each store has approximately 60,000 individual products, called stock keeping units (SKUs), on its shelves. Data is collected at several interesting places in a grocery store. Some of the most useful data is collected at the cash registers as customers purchase products. The point-of-sale (POS) system scans product barcodes at the cash register, measuring consumer takeaway at the front door of the grocery store, as illustrated in Figure 3-2's cash register receipt. Other data is captured at the store's back door where vendors make deliveries.



Allstar Grocery
123 Loon Street
Green Prairie, MN 55555
(952) 555-1212

Store: 0022
Cashier: 00245409/Alan

0030503347 Baked Well Multigrain Muffins	2.50
2120201195 Diet Cola 12-pack	4.99
Saved $.50 off $5.49	
0070806048 Sparkly Toothpaste	1.99
Coupon $.30 off $2.29	
2840201912 SoySoy Milk Quart	3.19
TOTAL	12.67
AMOUNT TENDERED	
CASH	12.67
ITEM COUNT: 4	

Transaction: 649 4/15/2013 10:56 AM

Thank you for shopping at Allstar

00649002204152013002454O9

Figure 3-2: Sample cash register receipt.

Source: Datawarehouse Toolkit 3rd Edition

Retail Case Study (contd.)

At the grocery store, management is concerned with the logistics of ordering, stocking, and selling products while maximizing profit. The profit ultimately comes from charging as much as possible for each product, lowering costs for product acquisition and overhead, and at the same time attracting as many customers as possible in a highly competitive environment. Some of the most significant management decisions have to do with pricing and promotions. Both store management and headquarters marketing spend a great deal of time tinkering with pricing and promotions. Promotions in a grocery store include temporary price reductions, ads in newspapers and newspaper inserts, displays in the grocery store, and coupons.

The most direct and effective way to create a surge in the volume of product sold is to lower the price dramatically. A 50-cent reduction in the price of paper towels, especially when coupled with an ad and display, can cause the sale of the paper towels to jump by a factor of 10. Unfortunately, such a big price reduction usually is not sustainable because the towels probably are being sold at a loss. As a result of these issues, the visibility of all forms of promotion is an important part of analyzing the operations of a grocery store. **Now that we have described our business case study, we'll begin to design the dimensional model.**